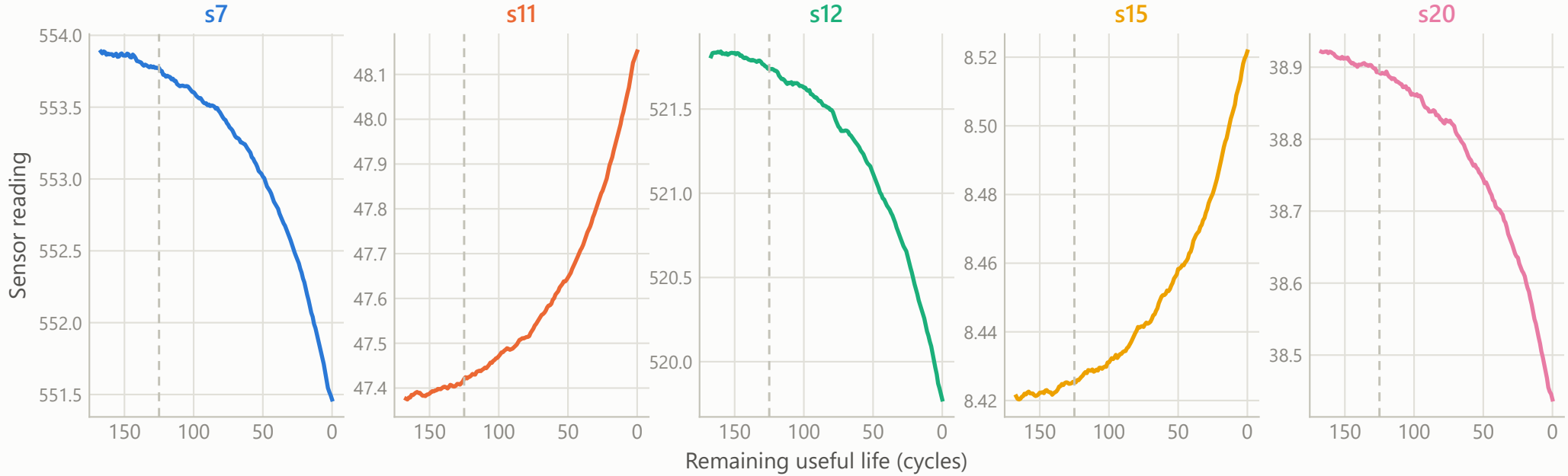
Every informative sensor drifts monotonically toward failure



Mean across 100 training engines (NASA CMAPSS FD001), smoothed over 7 cycles; each panel keeps its own raw scale. Failure is at RUL = 0, on the right of every panel.

Dashed line: the 125-cycle cap applied to training labels. The drift is gentle above it and steepens below.